

✦ PACK PLATE ✦

— Compatible campus dining for the whole pack —

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Mission Statement

NC State students with different dietary needs must search fragmented dining information one item at a time, making group meal decisions difficult. PackPlate combines campus hours, menus, nutrition, dietary filters, crowd levels, and transparent data warnings in one mobile-friendly view. We will test whether groups can choose a compatible dining location in under two minutes, compared with the current ten-minute process.



Before

- Audited an existing full-stack food application that the team did not write.
- Produced 20 evidence-backed use cases from the existing system.
- Ran identical prompts across 3 cloud LLMs and 1 locally hosted LLM
- Identified defects, including an allergen check that was not enforced on the server.
- Tested model reliability by recording incorrect, partial, and hallucinated answers.
- For project 1a, we have 59 test cases, with 44 passing and 15 failing.



Milestones

Now

- Build a responsive PackPlate prototype showing NC State dining locations, hours, menus, and nutrition data.
- Implement dietary filtering with fits, conflicts, and data incomplete results.
- Add multi-person filtering so a group can compare several dietary profiles against the same locations.
- Add simple student crowd reports using a small set of choices such as not busy, moderate, and crowded.
- Test the filtering logic against a hand-labeled menu dataset and evaluate whether student groups can choose a location in under two minutes.



Future

- Use class schedules and walking distance to suggest dining locations during schedule gaps.
- Add coordination features for groups, such as shared meal links or voting.
- Use accumulated reports to estimate busy periods by location and time.
- Improve ingestion reliability and support additional data sources or campus locations.
- Evaluate whether a native mobile application is justified after testing the web prototype.



Stakeholders

- **NC State students** – Need fast, accurate information when choosing where to eat.
- **Students with allergies** – Could be harmed by incomplete or incorrect allergen information.
- **Students with religious or dietary restrictions** – Need information about halal, vegetarian, vegan, and other requirements.
- **Groups of friends** – Need to find one location that works for several people simultaneously.
- **NC State Dining** – Owns or publishes the underlying menu, nutrition, and hours data.
- **Dining-hall staff** – May receive questions or complaints when published information is incomplete.
- **NC State administrators** – Care about student services, accessibility, privacy, and institutional reputation.
- **PackPlate maintainers** – Must monitor stale data, scraper failures, crowd reports, and deployment issues.
- **University legal/privacy staff** – Need the product to avoid unsafe claims and unnecessary storage of dietary information.



QR Codes



[Github Repo](#)



[Discord](#)



Tech Stack

